ZCAS University

CIT1142 EMERGING TRENDS IN ICT

MID SEMESTER EXAMINATION

FRIDAY 24TH APRIL 2026

10:30 – 12:00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Choose any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

QUESTION 1

A global research consortium consisting of several universities in all continents collaborates on climate modeling and disease outbreak prediction. Each university has its own computing resources, including servers and storage systems. Due to increasing computational demands, the consortium decides to implement a Grid Computing system that allows sharing and coordinated use of distributed resources across multiple institutions. The system integrates heterogeneous resources, supports large-scale data processing, and enables researchers to run complex simulations collaboratively.

Required:

(a) Define Grid Computing and explain how it differs from traditional distributed computing systems. **(5 marks)**

(b) Identify and explain three key characteristics of Grid Computing that make it suitable for the scenario above. **(5 marks)**

(c) Describe the architecture of a Grid Computing system, clearly explaining the role of each layer. **(5 marks)**

(d) Discuss two benefits and two challenges of implementing Grid Computing in the given scenario. **(5 marks)**

(e) Suppose a researcher submits a large simulation job to the grid:

i. Explain how resource allocation and scheduling would be handled in the Grid environment. **(2.5 marks)**

ii. Discuss the importance of middleware in this process. **(2.5 marks)**

[Total: 25 Marks]

QUESTION 2

A large retail company, Shoprite, operates multiple stores across Zambia. The company collects vast amounts of data from customer transactions, loyalty cards, and online purchases. Management wants to use Data Mining techniques to analyze this data to understand customer behavior, improve sales, and optimize inventory management.

Required:

(a) Define Data Mining and explain its role in knowledge discovery from large datasets. **(5 marks)**

(b) Identify and explain three common Data Mining techniques that could be applied in the scenario above. **(5 marks)**

(c) Describe the Data Mining process (KDD process), outlining each stage involved. **(5 marks)**

(d) Discuss two practical applications of Data Mining in the retail industry based on the scenario. **(5 marks)**

(e) A dataset from the company contains missing and inconsistent values:

i. Explain the importance of data preprocessing in Data Mining. **(2.5 marks)**

ii. Describe two techniques used to handle such data quality issues. **(2.5 marks)**

[Total: 25 Marks]

QUESTION 3

A financial institution, Zanaco, is modernizing its digital platforms to improve customer experience. The bank plans to develop a web-based service platform that allows customers to check account balances, transfer funds, pay bills, and access financial reports through web and mobile applications. To achieve this, the system will integrate multiple applications such as payment gateways, authentication systems, and third-party financial services using web technologies.

Required:

(a) Define Web Services and explain their role in modern computer applications. **(5 marks)**

(b) Identify and explain three key technologies or standards used in implementing web services. **(5 marks)**

(c) Describe the architecture of web services, clearly explaining how different components interact in the given scenario. **(5 marks)**

(d) Discuss two advantages and two limitations of using web services in the banking system described above. **(5 marks)**

(e) The bank needs to ensure secure and reliable transactions:

i. Explain the importance of security mechanisms in web services. **(2.5 marks)**

ii. Describe how APIs and middleware facilitate integration between different applications in this system. **(2.5 marks)**

[Total: 25 Marks]

END OF MIDSEMESTER EXAMINATION